Robot Project guidelines

Your project requires you to build it each week, not only the design itself, but also the report. In addition, you will be doing some data tracking, similar to what an engineering design team must do, where they track expenses, and time on a project.

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<u>Project Milestones:</u>

- send team member names to James (TA) by the end of 4 February (23:59) via email (james.achumu@nulondon.ac.uk)
- project title and team contract submitted by the end of 11 February (23:59)(formative)-submit on Canvas (group submission)-"MODULE: Robot Design Project/Teams contract". You mention these in the team contract: **Title of the project, Team members, Area of expertise for each member, Exact time you meet every week and duration of each meeting. The portal for submitting this document will be open after 4 February when you finalize your team members (this is a group submission)**
- project title discussed in the class 20/21 February
- proposal presentation (formative) 17/18 March (group submission)
- prototype presentation (summative) 31 March/1 April
- final project presentation (summative) 10/11 April

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<u>Time and Cost Accounting:</u>on the project in chronological order must be continuously maintained and updated throughout the project. It will be submitted in the final report. Each week of the project should be a separate row of the table, with separate columns to indicate the number of hours spent on the project by each teammate. The total time spent on the project by each teammate should be found on the bottom row of the table. Individual project grades may be adjusted to account for over- or under-participation, using both this information as well as mid- semester and end-of-semester peer evaluations.

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<u>Design project phases and report sections:</u>

Here are some suggestions on when these should be done:

Include these as part of the proposal and proposal presentation

Problem Definition

This starts with a clear problem statement. This includes goals, later used in your decision analysis. Functions, objectives and constraints that guide your design. Remember tools such as Duncker Diagrams and fishbone diagrams to help in this stage.

Alternative Designs

Generate many designs using creativity techniques as discussed last semester and in the Top Hat text. Remember that this is more than brainstorming, use another technique (SCAMPER, Biomics, Inversion, C-skectch or Gallery), and include sketches with short descriptions in the report. Consider FOC as you look at each, but remember, quantity here.

Decision Making

Use your goals in a rank order matrix, then use Decision Matrix. This can be used on different sections, and more than once. Use excel, include the matrices in the report along with use your goals in a rank order matrix, then use Decision Matrix. This can be used on different sections discussion of your process and choices.

Implementation

This is where you include the bill of materials and costs for the project, discussion of materials used, how it works, etc. An update about your project.

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Important notes:

- The prototype should do at least 75% functionality of the full artefact. So you break down the task to at least (and ideally) 4 subtasks and when you present your prototype, at least 3 of them should be accomplished by your design.
- The report has two parts. The first part is about the design and the second part is is related to ethics. In this part students should choose a general topic related to their design. For example, if the artefact of a group was a robot which can move and collect samples in hazardous environments, student can choose nuclear power plants, which have some hazardous environments, as the topic for this second part of their report. Students must analyse the topic ethically using the ethical theories they learned during the semester. **<u>Consider these parts of your report when defining your design project topic.</u>** You will find more details in AE1 brief.

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Grades:

The baseline grade for the Robot project will be approximately done as follows:

50% prototype presentation (group)

20% final oral presentation (group)

30% final written report (individual)

Each member must write their own final report.